ISYS2120 Sem2, 2025

Assignment 2 Group Report

Group code:

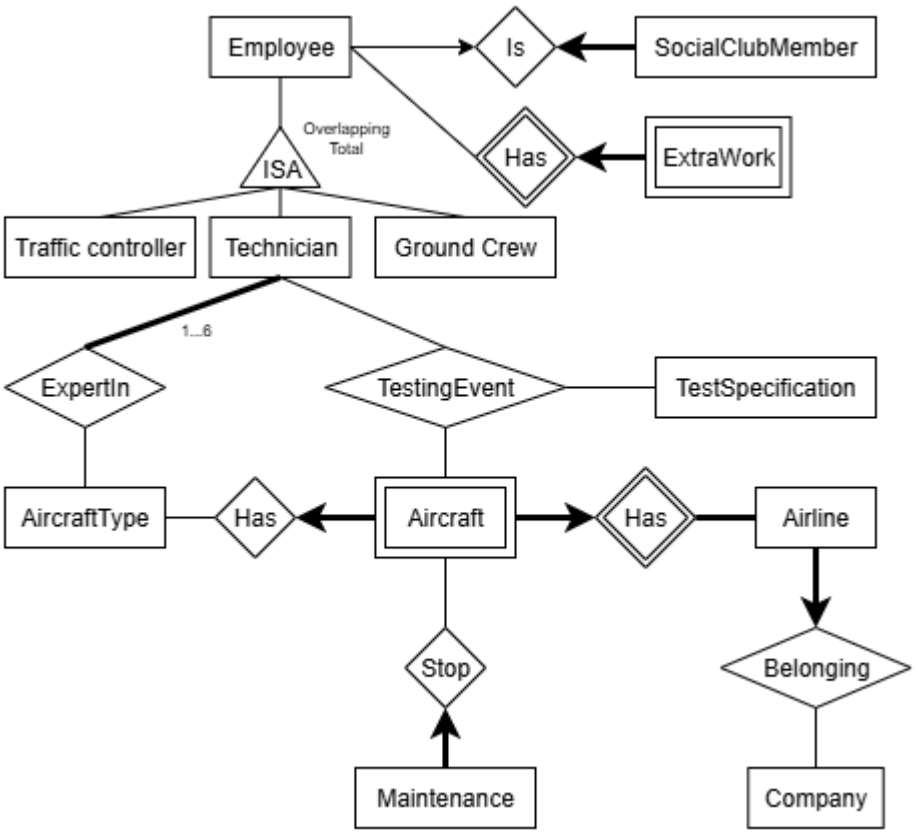
CC10-Team 03

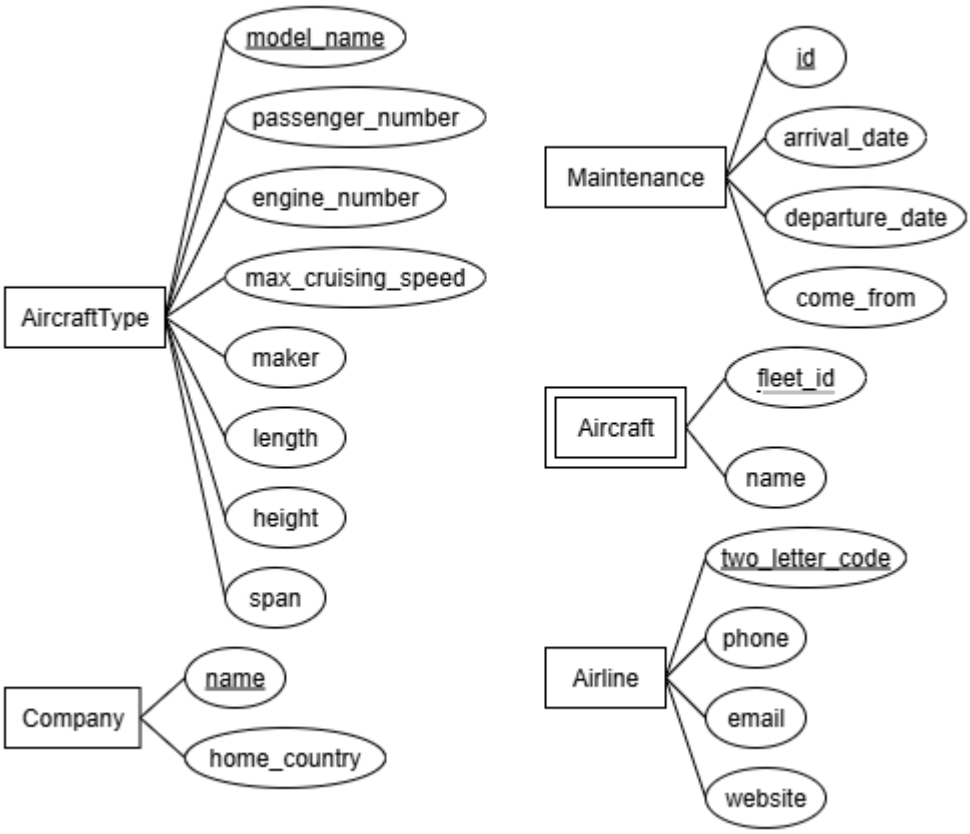
Group members:

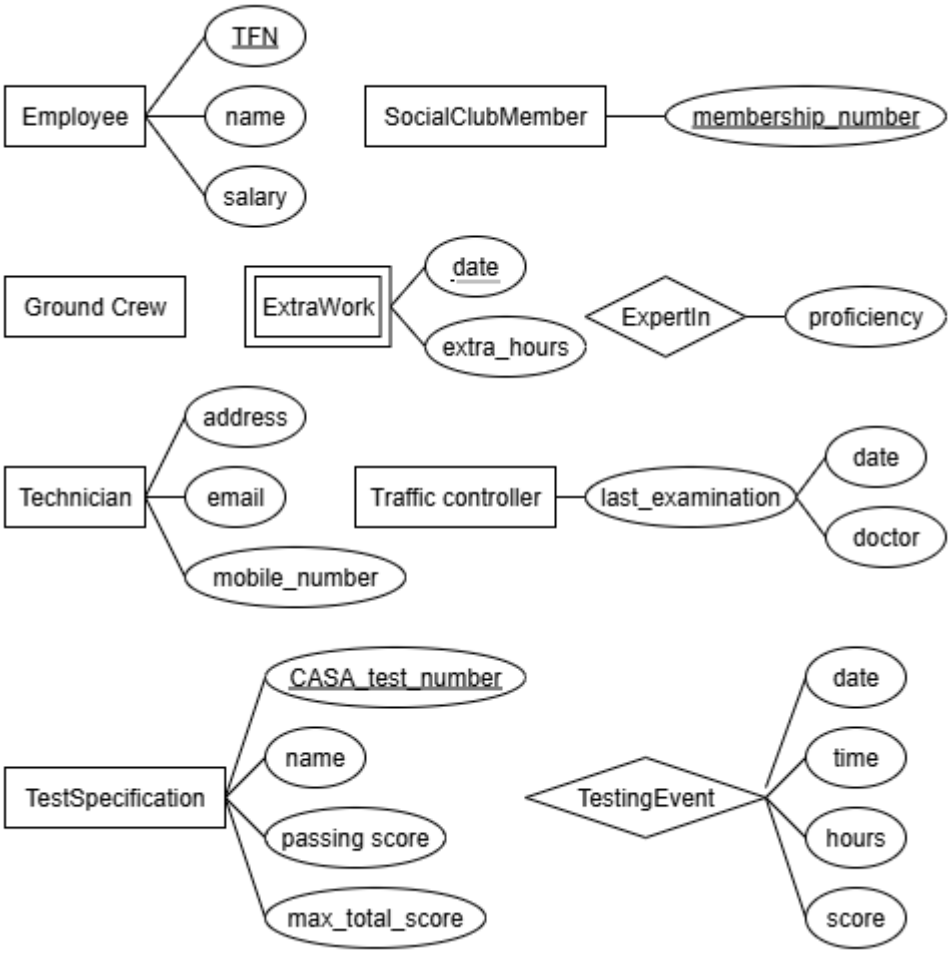
Haochen Liu SID: 540243105
Tingrui Chen SID: 540233036

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Part A


Employee
Is
SocialClubMember
Overlapping
Total
ISA
Has
ExtraWork
Traffic controller
Technician
Ground Crew
1...6
ExpertIn
TestingEvent
TestSpecification
AircraftType
Has
Aircraft
Has
Airline
Stop
Belonging
Maintenance
Company


model_name
passenger_number
engine_number
max_cruising_speed
AircraftType
maker
length
height
span
id
arrival_date
Maintenance
departure_date
come_from
fleet_id
Aircraft
name
two_letter_code
phone
Airline
email
website
name
Company
home_country


Employee
TFN
name
salary
SocialClubMember
membership_number
Ground Crew
ExtraWork
date
extra_hours
ExpertIn
proficiency
Technician
address
email
mobile_number
Traffic controller
last_examination
date
doctor
TestSpecification
CASA_test_number
name
passing score
max_total_score
TestingEvent
date
time
hours
score

Part B:

Here are the assumptions we made in our ER Diagram Design:

First we assumed that a **Company** may exists even if it does not own any **Airlines**. So that's why we drew a thin line with no arrows on the Company side of the **Company–Airline** relationship. Also in **Airline-Aircraft** relationship, we assumed that each **Airline** has at least one **Aircraft** as an airline cannot meaningfully exist without at least one aircraft in its fleet.

In the **Aircraft-Maintenance** relationship, we assumed that an Aircraft can have no maintenance events (i.e. it is newly registered at the airport), so we represented it by putting a single thin line on Aircraft side of the **Aircraft-Maintenance** relationship.

Our model also allows for an **AircraftType** to be recorded in the database without any associated **Aircraft**. This reflects the possibility that certain aircraft types may exist in general, but not have appeared in the airport. We used a single line to represent that on the **Aircraft side** of the **AircraftType-Aircraft** relationship.

For the **Employee** Entity, we think that an **Employee** can belong to more than one of the three roles (**GroundCrew**, **Technician**, **TrafficController**) and belongs to at least one. We expressed that by applying a notation of "overlapping total" constraint on the **Employee** entity.

For the **Aircraft** Entity, we didn't treat it as a sub-entity. Instead, we treated it as a weak entity of **Airline.** We decomposed the RegistrationNo into two parts. The first part is the two-letter airline code, which we stored in the **Airline** entity. The second part is the four-letter fleet id, which we stored it in the **Aircraft** entity because the specification explains that each registration number is made up of a code that identifies the airline owner and a unique identifier of airline's fleet. Here we define the "aircraft owner" as **Airline**. By separating the registration number into these two components, our model highlights the dependency of aircraft on airlines.

Strength:

All entities from the specification are represented in the ER Diagram. This ensures the accuracy and completeness of the conceptual description. Also, we have also explicitly expressed the multiplicities of relationships in our ER diagram by using appropriate notations such as thick or thin lines or arrows. Simultaneously, our model emphasize some key constraints mentioned in the requirements. For example, we documented the rule that each technician must be an expert in between one and six aircraft types by annotating this constraint '1..6' in the ER diagram. Lastly, our ER diagram effectively uses specialization by modeling the **Employee** entity with an ISA relationship into the subtypes **GroundCrew, Technician, and TrafficController**.The specialization is annotated as "overlapping and total", which conveys that each employee must belong to at least one subtype, while also allowing employees to hold multiple roles if required.

Limitations:

Some of the relationships cannot be expressed in the ER diagram. For instance, the ER diagram cannot explicitly express which role combinations the **Employee** can be when an employee belongs to multiple subtypes (such as whether Employee can be **Technician** and **TrafficController** or can be **Technician** and **Ground Crew** or anything else).

Also, the model is designed with a single airport (WSA). If the system were to be extended to multiple airports, additional entities and relationships would be required to capture the added complexity, which may influence scalability in its current form. For example, we need to add **Airport** entity to show multiple airports and also we need to add some relationships such as **Employee–Airport (worksAt), estingEvent–Airport (occursAt)** to handle specific airport activities.